Deep Learning-Enabled Overlapping Multiplexed Direct Absorption Spectroscopy for Synchronously Multicomponent Gas Concentrations Measurement

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***Abstract*—Direct absorption spectroscopy (DAS) offers the advantage of a simple system architecture for noncontact, real-time detection of gas concentrations. However, for multigas detection, existing DAS systems either rely on time-division multiplexed laser wavelength scanning, which inevitably results in asynchronous detection of different gases, or require that the target gases exhibit closely spaced or even overlapping absorption lines, thereby restricting the choice of laser wavelengths and limiting the range of detectable gas species. To overcome these challenges, we propose an overlapping multiplexed DAS (OM-DAS) scheme that synchronously scans the wavelengths of multiple lasers to probe multigas absorption information. Then, a novel end-to-end multigas analysis network (E2E-MAN) is proposed, which enables simultaneous spectral component separation and concentration regression of multiple gases. Experimental validation on a methane and carbon dioxide mixed gas system demonstrates that the proposed OM-DAS framework achieves high measurement accuracy, strong cross-interference suppression capability, and excellent long-term stability for synchronous multigas detection. Allan variance analysis shows minimum detection limits (MDLs) of 55.656 ppb for methane and 2.038 ppm for carbon dioxide at an averaging time of 300 s. These results indicate that the proposed deep learning-enabled OM-DAS provides a highly reliable, low-cost, and low-complexity method for synchronously detecting multicomponent gas concentrations.**

***Index Terms*—Deep learning, direct absorption spectroscopy (DAS), multicomponent gas concentration measurement, neural network, overlapping time multiplexing.**

I. Introduction

IN RECENT years, synchronous measurement of multicomponent gas concentrations has been urgently needed in fields such as combustion diagnostics, process control, and environmental monitoring [1], [2], [3]. In combustion diagnostics, synchronous measurement of combustion process gas concentrations is helpful in accurately determining the combustion state and improving combustion efficiency [4], [5]. In process control, synchronous measurement of multicomponent gas concentrations can adapt to transient reaction processes, providing real-time feedback to the control system, thereby enhancing system stability and control accuracy [6]. Furthermore, in environmental monitoring, gas emissions are influenced by wind speed and turbulence intensity. Synchronous monitoring can enhance the accuracy of emission estimates, thereby improving the reliability of environmental assessments [7], [8].

Tunable diode laser absorption spectroscopy (TDLAS), a spectroscopic measurement technique, has been widely applied in multicomponent gas detection due to its advantages of fast response, high sensitivity, low detection limits, and noninvasiveness [9], [10], [11]. TDLAS can be generally classified into direct absorption spectroscopy (DAS) and wavelength modulation spectroscopy (WMS). In DAS, the gas absorption line is obtained by directly measuring the variation of transmitted intensity, whereas WMS measures absorption-induced harmonic signals by superimposing a high-frequency modulation on the laser wavelength. Compared to WMS, DAS features a simple system structure, which has garnered extensive attention in recent years [12], [13], [14].

However, typical DAS systems are designed for the detection of single-component gas due to the limited wavelength tuning range [15]. To enable multicomponent gas detection, a mainstream scheme is to combine DAS with time-division multiplexing (TDM) technology, in which different laser wavelengths or multiple laser sources are switched sequentially in different time slots to measure the concentrations of different target gases separately [16]. Xu et al. [17] combined DAS with TDM, using a mechanical optical switch to control the timing of distributed feedback (DFB) lasers at different wavelengths. This setup enabled sequential measurement of methane [CH_4] and carbon dioxide [CO_2] concentrations. Besides, combining multiple photodetectors with TDM for measuring the spectral line of different gases is a viable approach. Dong et al. [18] develop a TDM-based multigas measurement system by using a single broadband light source. Although these methods offer advantages such as system simplicity and low signal processing algorithm complexity, the measurements of the different components are not synchronous.

In addition to the TDM-DAS scheme, another scheme is to utilize absorption lines that are inherently close to or

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even overlapping, employing a single laser for wavelength scanning. Synchronous multicomponent concentration measurement can then be achieved through typical multivariate regression methods such as principal component analysis (PCA), partial least squares (PLS), and nonlinear least squares (NLS). Although the multivariate regression methods can address cross-interference problems in spectral analysis to some extent [19], [20], [21], they typically rely on distinct spectral features and careful preprocessing, and their performance is often sensitive to noise and baseline drift. Therefore, they often exhibit low concentration regression accuracy under highly overlapping and dynamic detection scenarios [22].

In recent years, the rapid development of deep learning technology has made it possible to overcome many limitations of traditional multivariate methods. Deep neural networks, by learning complex nonlinear relationships from spectral data, exhibit stronger robustness in highly overlapping and dynamic detection scenarios. Sun et al. [23] utilized overlapping absorption lines of target gases and introduced a neural separation network to directly determine the gas concentration of each component; however, the absence of intermediate spectral outputs limits the physical interpretability of the model. To improve interpretability, Xiong et al. [24] developed a TDLAS-based multicomponent gas detection sensor, which employs an end-to-end aliased spectrum decoupling algorithm to separate the overlapping absorption lines and determine concentrations. While these methods that combine inherent overlapping spectra with deep learning have achieved promising results for multicomponent gas concentration measurement, their performance is constrained by the fixed wavelengths of specific absorption lines and the types of target gases, preventing their potential to fully achieve high sensitivity and accuracy.

In this work, a novel overlapping multiplexed DAS (OM-DAS) detection scheme is proposed. In this scheme, the high-sensitivity absorption lines of target gases are intentionally overlapped by synchronously coupling multiple DFB lasers operating at specific wavelength bands. An end-to-end multigas analysis network (E2E-MAN) is employed to achieve overlapping spectral component separation and concentration regression. The proposed E2E-MAN model unifies baseline removal, spectral decomposition, and concentration prediction tasks within a single network framework. Deep residual multilayer perceptions are used to disentangle the transmission spectra of individual gases from complex overlapping spectra with a hybrid baseline, while subsequent residual-connected layers perform concentration regression. A transfer learning strategy is adopted, where the spectral separation module is pretrained and subsequently optimized jointly with the full model, allowing the network to directly learn the mapping from overlapping spectra to quantitative gas concentrations. This not only accelerates the training convergence speed but also significantly improves the accuracy and stability of concentration regression. In the end, the E2E-MAN model was experimentally validated on both simulated and real experimental datasets obtained from a CH_4/CO_2 mixed-gas system. The results obtained demonstrate that the proposed method eliminates the temporal delay inherent in TDM-DAS systems and overcomes the reliance on inherently overlapping absorption lines, thereby enabling low-cost and synchronous multigas concentration detection with promising potential for generalized gas analysis applications.

The structure of this article is as follows. Section II presents the design details of the OM-DAS experimental setup. Section III describes the principle of OM-DAS, dataset construction, implementation details of the network model, and training optimization strategies. Section IV discusses the spectral separation performance and concentration regression results. Finally, Section V concludes this article.

II. Sensor Configuration

A. Selection of Absorption Lines

The rational selection of the target gas absorption lines is the key to ensuring the sensitivity and selectivity of the system [25]. Typically, the selection is based on line strength, potential interference from other gases, and the availability of lasers in the corresponding spectral band [26], [27]. The test gases (CH_4 and CO_2) exhibit significant absorption characteristics in the near-infrared band. As shown in Fig. 1, based on the parameters of the HITRAN database, the absorption spectra of CH_4 and CO_2 in the wavenumber range of 6030–6380 cm^{-1} were simulated. Obviously, the regions around 6047 cm^{-1} for CH_4 and 6360 cm^{-1} for CO_2 satisfy the requirements for detection wavelengths.

B. Experimental System Configuration

The experimental system consists of an optical detection module, a signal acquisition and processing module, and a gas mixing module. Fig. 2 presents the hardware architecture of the system. The optical detection module employs two DFB lasers based on OM-DAS. One laser has a central wavelength of 1572 nm (tuning range 1571–1573 nm, output power 10 mW, current tuning coefficient 0.01 nm/mA, and linewidth 3 MHz), and the other has a central wavelength of 1653 nm (tuning range 1652.7–1654.7 nm, output power 10 mW, current tuning coefficient 0.01 nm/mA, and linewidth 2 MHz). The two beams are coupled into a gas cell by an optical coupler. Finally, the photodetector (PDA05CF2, Thorlabs) converts the transmitted light into an electrical signal. In the signal acquisition and processing module, the data acquisition card (USB-6361, NI) synchronously drives the laser tuning via sawtooth signals and records the photodetector output. In the gas mixing module, the mass flow controller (MFCD519MG) blends standard gases following a preset concentration script. All experiments in this article were performed under laboratory conditions at ambient temperature and pressure (297 K, 1 atm). The pressure inside the gas cell was monitored using a rear-mounted pressure gauge. The premixed gas was introduced into the evacuated cell until the internal pressure stabilized at 1 atm. The temperature was maintained at approximately 297 K using indoor air conditioning. Small errors and fluctuations in temperature and pressure produced only minor variations in absorption line intensity and shape, which were significantly smaller than the changes induced by variations in gas concentration [28].

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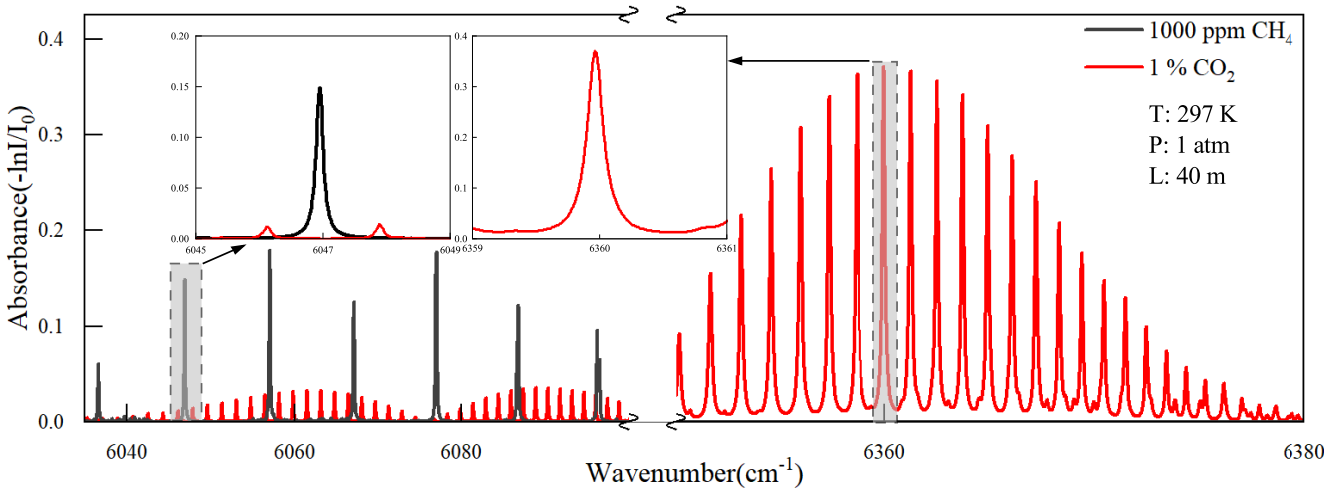


Fig. 1. Absorption spectra of CH_4 and CO_2 in selected near-infrared regions, with the inset highlighting the chosen absorption lines.

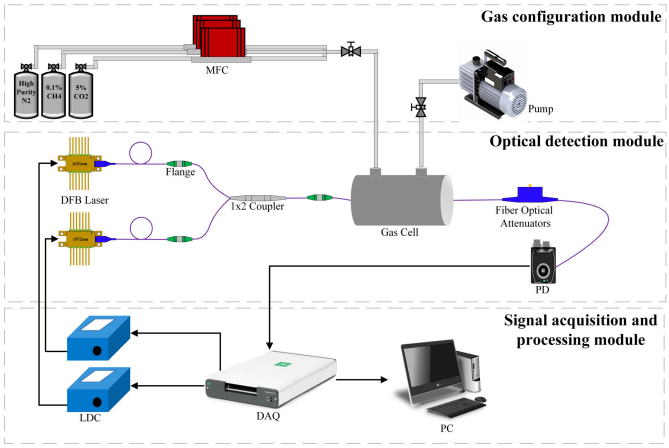


Fig. 2. Schematic of OM-DAS experimental setup. MFC: mass flow controller, LDC: laser diode controller, DFB Laser: distributed feedback laser, PC: personal Computer, and DAQ: data acquisition card.

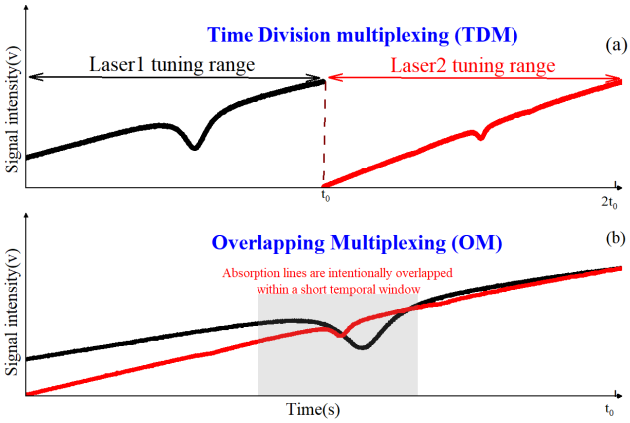


Fig. 3. Comparison of traditional TDM and OM detection schemes (a) traditional TDM and (b) proposed OM.

III. Methodology

A. OM-DAS Principle

Building on the core idea of multicomponent gas synchronous detection, this section examines the principle of OM-DAS and deconstructs the real experimental spectra. Traditional TDM-DAS schemes achieve multigas detection by sequentially assigning time slots to different lasers, thereby retrieving gas concentrations based on the Beer–Lambert law [29]. However, this scheme introduces temporal delays between the absorption lines of target gases, thereby lacking the capability of synchronous detection. On the contrary, the proposed OM-DAS scheme employs multiple lasers to synchronously scan the gas mixture, forcing the absorption lines of target gases to overlap at the same moment, which enables synchronous multicomponent gas detection while maintaining system simplicity and low cost. Fig. 3 compares the principles of traditional TDM and the proposed OM-DAS.

Under the OM-DAS detection scheme, by appropriately selecting gas absorption lines, the ith laser is only used for measuring the absorption line of the ith target gas (other gases have almost no absorption). Since there is no wavelength overlapping among m lasers in synchronous scanning, the light intensities of different wavelengths λ are directly superimposed. Therefore, based on Beer–Lambert's law, the measurement process can be expressed as follows:

$$I(\lambda) = \sum_{i=1}^{m} I_i(\lambda) = \sum_{i=1}^{m} I_{0,i}(\lambda)\, e^{-\alpha_i(\lambda) C_i L} \tag{1}$$

where $I_{0,i}(\lambda)$ and $I_i(\lambda)$ are the light intensities before and after transmission from the ith laser, $I(\lambda)$ is the sum of the light intensities after transmission from all lasers, $\alpha_i(\lambda)$ and C_i are the absorption coefficient and concentration to be measured of the ith gas, respectively, and L is the absorption optical path. Because the laser scanning speed is constant and much slower than the response of the photodetector, the detected voltage signal has a linear relationship with the light intensity [30]. $I_i(\lambda)$ is linearly mapped from the optical wavelength domain to the time domain, resulting in the time-domain signal $B_i(t)T_i(t)$

$$U_i(t) = B_i(t)\, T_i(t) + n_i(t) \tag{2}$$

where $U_i(t)$ denotes the photodetector signal of a single laser, $n_i(t)$ represents measurement noise, which originates from factors such as detector noise, laser frequency jitter,

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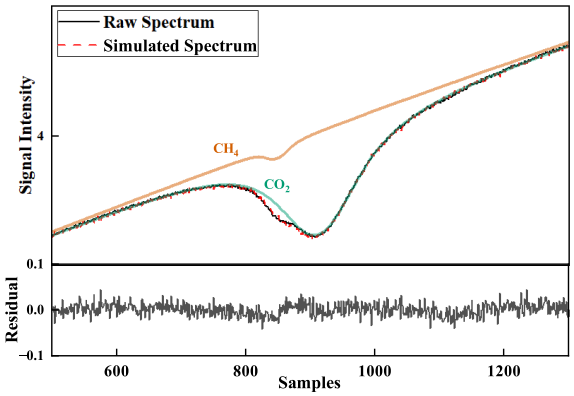


Fig. 4. Comparison of a representative experimental measurement with a simulated overlapping spectrum with overlap coefficient η of 0.1. The translucent orange and cyan lines indicate the individual components that contributed to the overlapping spectrum.

and environment interference, which usually follows a Gaussian statistical distribution [31]. $B_i(t)$ and $T_i(t)$ represent the baseline and the transmittance, respectively. The baseline was modeled using a first-order polynomial to reproduce the linear power variation of the laser diode under sawtooth-wave modulation, while random fluctuations were introduced to simulate thermal drift during long-term measurements [32]. The total measured signal $U(t)$ can be derived as

$$U(t) = \sum_{i=1}^{m} U_i(t) = \sum_{i=1}^{m} B_i(t)\, T_i(t) + n(t). \tag{3}$$

The baseline and transmission spectrum of each laser photodetector signal are multiplied together and then superimposed on other laser detection signals, jointly influencing the final overall baseline and transmission spectrum signal. Furthermore, the degree of overlap between adjacent absorption lines i and j can be controlled by adjusting the voltage tuning range of each laser. The overlap is quantified by an overlap coefficient $\eta_{i,j}$, which measures the fraction of the spectral area shared by the two absorption lines relative to their total areas, and is defined as

$$\eta_{i,j} = \frac{\int \min\left(-\ln\left(T_i(t)\right), -\ln\left(T_j(t)\right)\right) dt}{\int -\ln\left(T_i(t)\right) dt + \int -\ln\left(T_j(t)\right) dt}. \tag{4}$$

B. Dataset

Since acquiring labeled OM-DAS experimental data is time-consuming and labor-intensive, a simulation dataset was constructed to train the deep neural network, and the trained model was subsequently evaluated in an experimental environment.

In the simulated dataset, the CH_4 and CO_2 concentration ranges were set to 1–100 ppm and 1 %–5 %, respectively, based on the natural gas pipeline leak detection scenario. Ideal transmission spectra generated from HITRAN were used as spectral labels, and a total of 20 000 overlapping spectral signals were synthesized according to (3). To make the simulated signal more closely resemble a real experimental signal, Gaussian white noise with a mean of 0 and a standard deviation of 0.0028 V, together with a baseline perturbation term with an amplitude of 0.00016 V, was introduced into the ideal transmission spectrum to construct the simulated spectral signal based on (3). Both parameters were estimated from a statistical analysis of the experimental signals, which typically range from 3 to 5 V. As shown in Fig. 4, the simulated signal closely matches the experimental signal, exhibiting low residuals. The dataset was divided into training and validation sets at a 4:1 ratio. To further assess model generalization, an additional 1000 overlapping spectra were generated with doubled noise standard deviation as a simulated test set.

To verify the model's generalization ability, an experimental dataset was constructed to evaluate the model's performance from three aspects. First, 1000 overlapping spectra were acquired at fixed concentrations of 60 ppm for CH_4 and 0.91 % for CO_2 to assess the spectral separation performance of the model. Second, experiments were conducted under a dynamic condition to evaluate the model's robustness, where CH_4 was maintained at 100 ppm while CO_2 varied in 1 % increments, and a total of 592 overlapping spectra were acquired in real time. Third, to evaluate the long-term stability of the E2E-MAN model, a total of 3600 overlapping spectra of a mixed gas containing 60 ppm CH_4 and 0.91 % CO_2 were continuously acquired in the laboratory. The true gas concentration value is calculated using the flow rate ratios of each MFC channel and the concentration of the standard gas cylinder.

C. Model Structure

The proposed deep learning model, E2E-MAN, incorporates attention mechanisms and residual connections. It is divided into two functional parts according to the processing tasks, which are responsible for spectral component separation and concentration regression. As illustrated in Fig. 5, the end-to-end model adopts a two-stage training strategy consisting of a spectral separation module and a concentration regression module, enabling seamless integration of baseline removal, spectral decomposition, and concentration prediction tasks within a unified deep learning pipeline.

In the spectral separation module, the front-end of the spectral attention block employs a multilayer perceptron (MLP) integrated with skip connections to adjust the feature responses across the entire spectral dimension. The back-end achieves feature fusion through a fully connected layer, thereby outputting spectral features embedded with the attention mechanism. To further enhance feature representation capability and preserve long-range dependencies in the spectrum, the attention outputs are fed into nine stacked residual blocks. Each residual block consists of two linear layers equipped with LayerNorm and the GELU activation function, as LayerNorm preserves interframe spectral consistency under baseline drift, with skip connections introduced to facilitate gradient flow, alleviate the gradient vanishing problem during training, and thus improve the accuracy of spectral separation. Notably, due to the overlapping transmittance spectral frame difference caused by baseline drift, the LayerNorm incorporated in the module plays a pivotal role in accelerating model training and enhancing model performance and generalization ability. Eventually, this module successfully removes the hybrid baseline and separates the independent transmission spectra corresponding to each component, which are then

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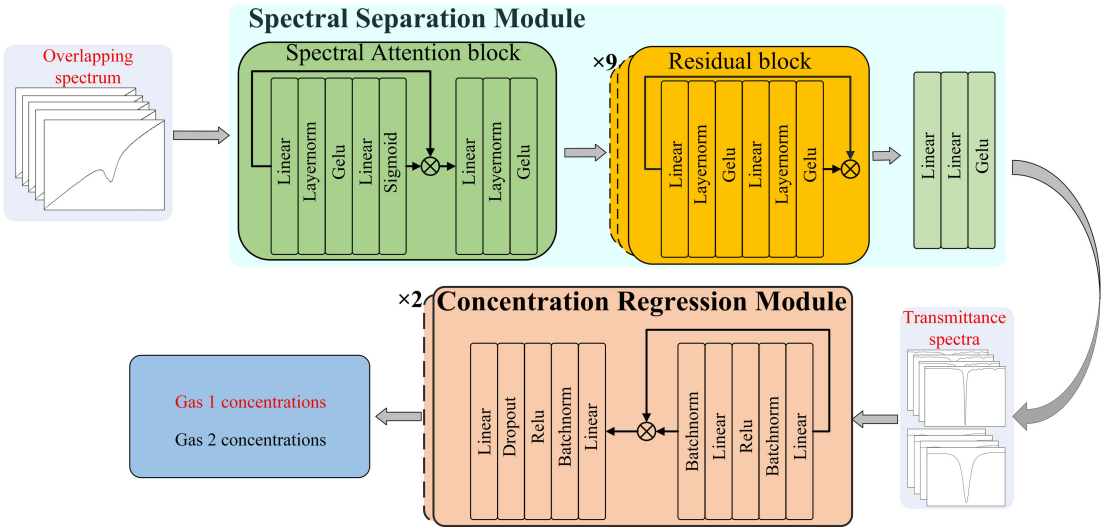


Fig. 5. E2E-MAN architecture diagram for OM-DAS.

fed as inputs to the downstream concentration regression task. For the concentration regression task, two dedicated concentration regression modules are designed to effectively maintain the independence of each gas absorption feature. Within these modules, the gas transmission spectra undergo linear transformation, BatchNorm, and ReLU activation layers in sequence for concentration feature extraction. A residual-like connection structure is also introduced to retain global information. After feature fusion and further processing, the concentration regression results are finally output. Unlike end-to-end networks that solely perform concentration regression, E2E-MAN incorporates intermediate output layers with explicitly defined physical meanings, which enable traceable decision-making processes, verifiable intermediate results, and identifiable error sources.

D. Model Training Process

In addition to an effective algorithmic model architecture, a well-designed training strategy is also crucial. In this work, the entire training process encompasses three key components: data generation, model training, and model evaluation.

During the data generation stage, each input overlapping spectrum is independently normalized on a per-sample basis. The ideal transmittance spectral labels and concentration labels are, respectively, normalized to the range between 0 and 1 for each gas component to enhance training stability and regression accuracy.

The model training adopts a two-stage training framework based on transfer learning. First, the spectral separation module is independently pretrained to establish a robust baseline for resolving overlapping spectra. Subsequently, this module is integrated into the complete architecture for joint end-to-end optimization. Significantly, all network parameters remain unfrozen, allowing gradients to propagate from the final output through to the initial input layers for comprehensive fine-tuning. For the distinct training tasks at different stages, this study designs a differentiated loss function structure. The loss function in the pretraining stage is composed of a weighted combination of multiple terms, including a coarse peak-region weighted mean square error (mse), first-order differential mse, and a Pearson correlation coefficient term between components. The coarse peak-region weighted mse enhances the model's focus on the characteristic absorption regions of CO_2 and CH_4 without requiring precise calibration of absorption peak positions. Pearson correlation loss ensures their independence by suppressing the covariance between the separated spectra. The first-order differential mse constrains the rate of change in the predicted spectra, effectively suppressing high-frequency noise and improving the smoothness of the reconstructed signals. The correlation coefficient term serves to reduce the correlation between the separated transmittance of different components, thereby mitigating cross-component interference. These three loss components are combined via empirically tuned weighting coefficients to achieve a favorable balance between separating accuracy and model stability. In the fine-tuning stage, the loss function is defined as the sum of the concentration prediction mse and the mse between the reconstructed overlapping spectra from the separated transmission spectra and the input spectra. This design allows minimizing the deviation between predicted and true concentration values while ensuring the spectral separation performance.

Finally, the trained E2E-MAN was evaluated on simulated and experimental data to assess its performance for mixed gas analysis. The overall training workflow is illustrated in Fig. 6. During the training process of the two models, we use a batch size of 32 and an Adam optimizer with an initial learning rate of 10^{-3}. The learning rate adjustment adopts the ReduceLROnPlateau scheduler. Each model is trained for a total of 300 epochs. In addition, our models run in the PyTorch framework (version 1.10.0) and Python 3.9 programming environment on the NVIDIA Tesla V100S-PCIe GPU platform, which provides sufficient computational capability for large-scale training and inference in typical TDLAS measurement systems.

E. Evaluation Indictors

Here are the evaluation indicators used in this article, as illustrated in Table I, where m represents the number of

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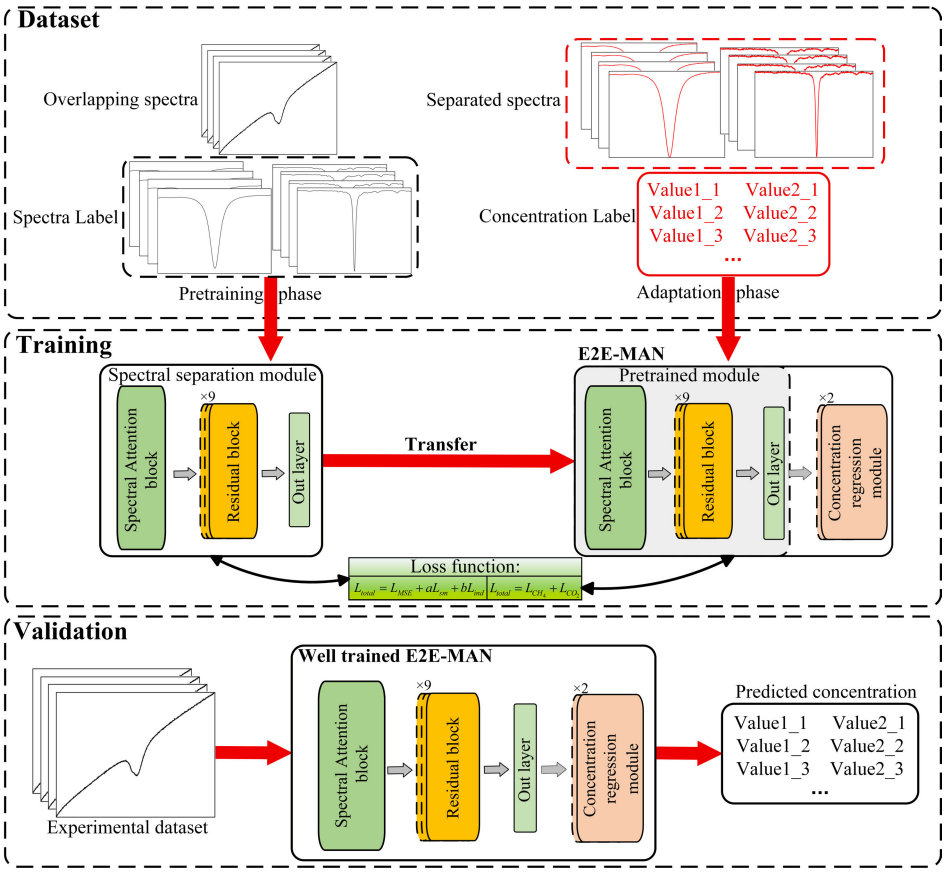


Fig. 6. Training process flowchart: the top box represents the dataset generation stage; the middle box denotes the model training stage; and the bottom box corresponds to the validation stage on experimental data.

TABLE I
EVALUATION INDICATORS AND THEIR FORMULAS

Evaluation indicator	Formula
Root Mean Square Error	$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(s_i - \hat{s}_i)^2}$
Absolute Error	$\mathrm{AE} = \frac{1}{m}\sum_{i=1}^{m}\lvert c_i - \hat{c}_i\rvert$
Relative Error	$\mathrm{RE} = \frac{1}{m}\sum_{i=1}^{m}\frac{\lvert c_i - \hat{c}_i\rvert}{c_i}$
Coefficient of Determination	$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{m}(c_i - \hat{c}_i)^2}{\sum_{i=1}^{m}(c_i - \bar{c}_i)^2}$

different components, n represents the number of spectral data points, c_i and $\hat{c}_i$ denote the true and predicted concentration of the ith sample, s_i and $\hat{s}_i$ denote the reference and measured spectra at the ith component, and $\bar{c}$ is the mean of true concentrations. The true concentration and reference spectra are obtained through labeling. In particular, the closer R^2 is to 1, the better the overall concentration prediction result of the method, and the smaller the AE and RE, the more accurate the concentration prediction result. The smaller the RMSE, the better the overlapping spectral component separating effect of the method.

IV. RESULTS AND DISCUSSION

To comprehensively assess the effectiveness of the algorithm, we validated the E2E-MAN model's performance in spectral component separation and quantitative concentration regression across multiple experiments with diverse characteristics. Moreover, a set of evaluation metrics was established to quantitatively assess the performance of the E2E-MAN. The performance of spectral component separation was assessed using the RMSE between the reconstructed overlapping spectra, which were generated by recombining the separated spectral components, and the original overlapping spectra. Concentration prediction was evaluated using R^2, AE, and RE.

A. Spectral Component Separation Performance Analysis

The spectral separation module learns to extract individual transmission spectra from the overlapping spectra, which provides a physically meaningful spectral representation for the subsequent concentration regression task; therefore, rigorous validation of the E2E-MAN is essential, particularly under low-concentration conditions where weak absorption features blur spectral overlaps. Fig. 7(a) shows the spectral separation results under these conditions (60 ppm CH_4 and 0.91 % CO_2). The RMSE values between the true and separated transmission spectra, as well as between the experimental spectrum and the baseline-reconstructed spectrum, were as low as 0.0000643, 0.000225, and 0.0017, respectively. Fig. 7(b) further shows the distribution of spectral reconstruction errors with Gaussian fitting, yielding a full width at half maximum (FWHM) of only 0.00196. To fully demonstrate the model's spectral component separation capability in continuous detection,

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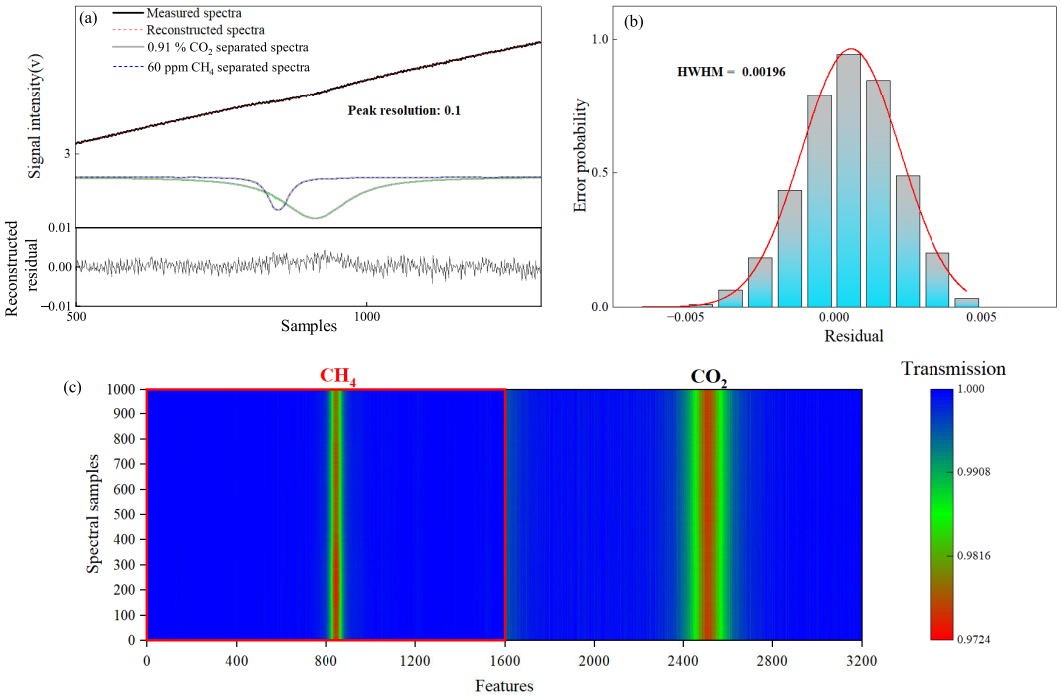


Fig. 7. Separated result of experimental overlapping spectrum. (a) Comparison between the separated transmission spectrum and the true transmission spectrum. (b) Distribution of spectral reconstruction errors and their Gaussian fitting results. (c) Separated transmission spectra contour map of 1000 consecutive experimental samples.

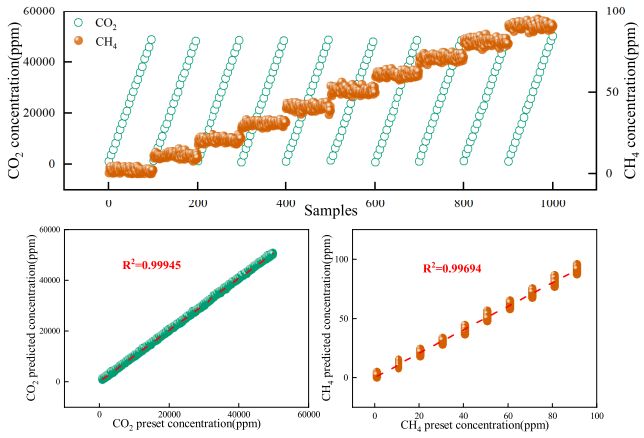


Fig. 8. Multicomponent gas concentration quantitative regression results of the E2E-MAN model for simulated spectra under strong noise interference, with CH_4 concentrations ranging from 0 to 100 ppm and CO_2 concentrations from 0 % to 5 %.

1000 consecutive separation results are presented, as visualized in Fig. 7(c). Along the spectral sample axis, the absorption valleys of CH_4 and CO_2 transmittance spectra are clearly resolved, with consistent color intensity and stable positional alignment, exhibiting no significant fluctuations. The results clearly demonstrate that E2E-MAN can achieve precise spectral component separation under weak spectral characteristics and high η (no less than 0.76), exhibiting strong stability and balanced performance.

B. Concentration Quantitative Regression Analysis

To further evaluate the performance of the E2E-MAN model from the perspective of quantitative gas concentration analysis, three experiments were set to evaluate the concentration prediction generalization, the robustness against cross-interference of different components, and long-term detection stability. Initially, the model's quantitative

TABLE II

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON SIMULATED TEST SET

Method	AE (mean)	RE (mean)	R^2	Calculation time
E2E-MAN	0.02% (CO_2) 1.25 ppm (CH_4)	1.86% (CO_2) 16.92% (CH_4)	0.9995 (CO_2) 0.997 (CH_4)	1.863 ms
NLS	0.18% (CO_2) 2.93 ppm (CH_4)	9.1% (CO_2) 25% (CH_4)	0.9862 (CO_2) 0.9993 (CH_4)	1.412 s

Note: The calculation times were measured on a PC with an 11th Gen Intel® Core™ i7-11700K CPU (8 cores, 3.60 GHz).

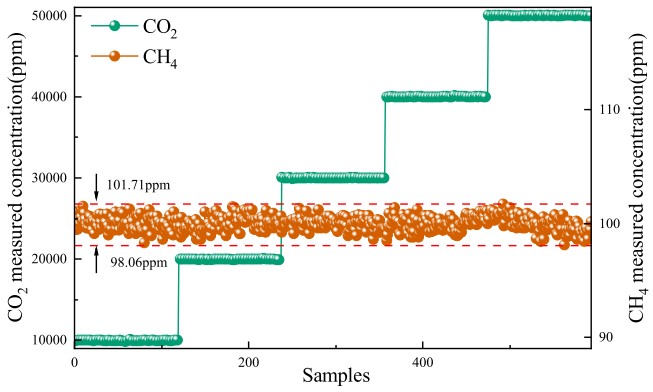


Fig. 9. Concentration prediction results under the condition where CH_4 is at 100 ppm and 0.91 % CO_2 varies dynamically from 1 % to 5 % in 1 % increments.

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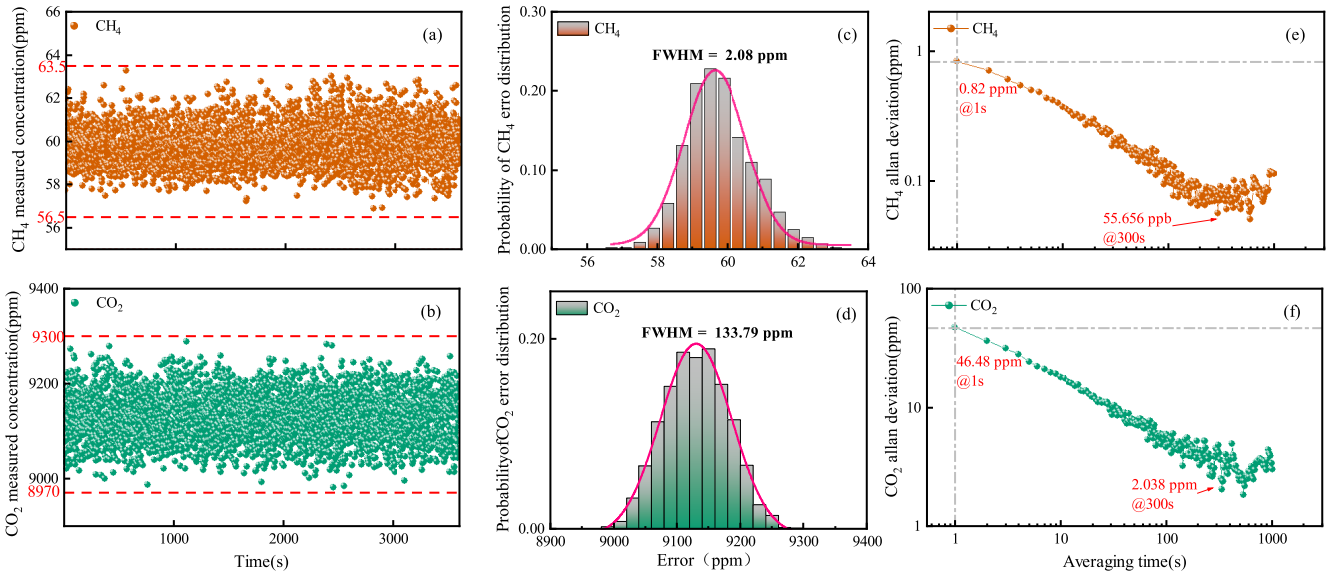


Fig. 10. Long-term concentration detecting results of CH_4 and CO_2 at fixed concentrations. (a) and (b) Concentration detecting results with preset CH_4 concentration at 60 ppm and CO_2 concentration at 0.91 %. (c) and (d) Error distribution histograms of the predicted concentrations for each component at their preset concentrations. (e) and (f) Allan deviation plots of the concentration monitoring result.

concentration regression capability was validated on simulated overlapping spectra with twice the original noise level across a wide concentration range. Fig. 8 presents the concentration prediction results in both overall and detailed aspects. Overall, the R^2 coefficients for CO_2 and CH_4 are 0.996 and 0.999, respectively. There is no significant fluctuation in the gradient AE and RE results for CO_2 and CH_4. The results prove that the model possesses adequate generalization capability to adapt to environmental variations. In addition, this article compares E2E-MAN with a traditional method based on NLS absorption line fitting to verify its performance advantage under multicomponent cross-interference. We conduct comparisons with the NLS method on a simulation test dataset. To ensure the accuracy of the experiments, the parameters of this method are optimized. Table II presents the mean AE, mean RE, R^2, and computation time of the two methods. The results show that E2E-MAN achieves higher accuracy in concentration regression. More importantly, its faster calculation time enables real-time concentration detection within the OM-DAS scheme, even at faster laser array scanning rates. Subsequently, to prove the model's robustness against crosstalk was validated in an experimental environment. We evaluated the robustness of the model to predict CH_4 concentration under varying CO_2 concentrations (100 ppm CH_4 and 1 %–5 % CO_2). Fig. 9 shows the predicted concentration results of the dynamic changes in the concentration ratios of the mixed gases. It can be directly observed that when CO_2 concentration changes rapidly, the CH_4 prediction remains unaffected and retains strong stability. Notably, regardless of environmental CO_2 variations, the prediction error for CH_4 stays within ± 2 ppm. These results sufficiently demonstrate that the features of each gas component exhibit strong independence.

Moreover, we continuously measured a mixed gas containing 60 ppm of CH_4 and 0.91 % of CO_2 for one hour to evaluate the long-term performance of the E2E-MAN model in the laboratory environment. Fig. 10(a)–(d) displays the concentration results of long-term monitoring of CH_4 and CO_2 at fixed concentrations. Compared with the true concentration values, the standard deviation of the predicted CH_4 concentration error is 0.95 ppm, while that of the predicted CO_2 concentration error is 49.91 ppm. Upon analyzing the concentration errors, the FWHM of the CH_4 concentration error distribution is 2.08 ppm, while for CO_2, it is 133.79.06 ppm. The Allan deviations of the two gases are shown in Fig. 10(e) and (f). The Allan deviation analysis indicates that, at an averaging time of 30 s, the minimum detection limits (MDLs) for CH_4 and CO_2 are 0.186 and 8.612 ppm, respectively. By further increasing the averaging time to 300 s, the MDLs can be reduced to 55.656 ppb for CH_4 and 2.038 ppm for CO_2. The variation in detection limits is primarily attributed to the different absorption line sensitivities of each gas in its spectral band. These promising results demonstrate the effectiveness of the proposed deep learning-enabled OM-DAS scheme, while also illustrating that our simulation method for model training is capable of capturing the underlying data distribution and generalizes well to actual experimental conditions.

Ultimately, as illustrated in Table III, we compared and analyzed the performance of our method with previously reported ones. The dual-gas sensor based on DAS reported in [23] achieved true simultaneous detection through synchronous scanning with a single laser, but it was limited by the inherent specific absorption line wavelengths and the categories of the measured components. Compared with the multicomponent gas detection systems based on the combination of TDM and DAS reported in [17] and [33], the proposed method eliminates the time delay introduced by TDM and achieves higher detection efficiency. Moreover, under multicomponent spectral interference, it provides higher detection accuracy than conventional TDM–DAS systems that have no multicomponent spectral interference. Overall, the approach proposed in this work enables synchronous multicomponent gas concentration

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TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED DETECTION METHOD WITH OTHERS REPORTED IN THE LITERATURE

Technology	Synchronicity	Wavelength(nm)	Optical length	Detection limit(gas)	Averaging time
OM-DAS (this work)	Synchronous	1653 and 1572	40m	0.5ppm (CH_4) 21.19ppm (CO_2)	7s
DAS with inherent overlap [23]	Synchronous	2330	20m	120.86ppm (CH_4) 0.5ppm (CO)	6.66s
TDM-DAS [17]	Asynchronous	3310, 4260 and 4650	393.2mm	2.84ppm (CH_4) 8.69ppm (CO_2) 8.83ppm (CO)	10s
Modified TDM-DAS [33]	Asynchronous	1653 and 1574	45m	$\sim$0.1ppm (CH_4) 35.97ppm (CO_2)	166s

detection in a more universal manner while maintaining a simple system architecture, which has not been achieved by other methods.

V. CONCLUSION

We propose a detection scheme based on OM-DAS and develop a deep learning model, E2E-MAN, to enable the synchronous acquisition of multicomponent gas absorption spectra and simultaneous concentration detection. Initially, the absorption spectra are obtained using an OM-DAS strategy, producing an overlapping spectrum for subsequent analysis. In the spectrum processing algorithm, the model removes the hybrid baseline, separates the transmission spectra, and predicts concentrations of different components from the overlapped spectrum. The proposed method demonstrated outstanding synchronous detection performance on a CH_4 and CO_2 mixed gas system. Under laboratory conditions, the system demonstrated excellent robustness against the cross-interference between CH_4 and CO_2, and the Allan deviation analysis from long-term detection experiments revealed MDLs of 55.656 ppb for CH_4 and 2.038 ppm for CO_2 at an averaging time of 300 s. Compared with traditional methods, the proposed approach provides a novel solution to the limitation of multicomponent gas detection capability in DAS systems, offering substantial potential for applications in portable spectroscopic sensing, industrial process control, and medical diagnostics.

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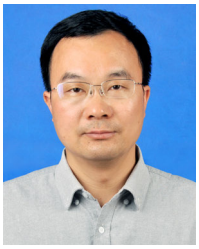
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